

GE AEROSPACE STOCK PREDICTOR

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ECEGR 5760

AGENDA

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Data & Model Architecture

2

Execution of the model

3

Compare performance w/
other stocks

Data & Model Architecture

Data & Preprocessing

- Dataset: Historical daily stock prices for GE, LMT, and XOM.
- Preprocessing:
 - Normalized with MinMaxScaler.
 - Converted into overlapping sequences (time_step = 20).
 - 80% for training, 20% for testing.

Model Architecture

- Simple RNN chosen for sequential data handling.
- 50 recurrent neurons to avoid overfitting.
- 2 RNN layers for pattern recognition.
- Dropout = 0.4 to prevent overfitting.
 - Epochs = 50, Batch size = 64.

```
# Load data
df = pd.read_csv('GE.csv')
df['Date'] = pd.to_datetime(df['Date'])
df.set_index('Date', inplace=True)

# Normalization
scaler = MinMaxScaler()
df_scaled = scaler.fit_transform(df[['Close']])

# Create Sequences
def create_sequences(data, time_steps=25):
    X, y = [], []
    for i in range(len(data) - time_steps):
        X.append(data[i:i + time_steps])
        y.append(data[i + time_steps])
    return np.array(X), np.array(y)

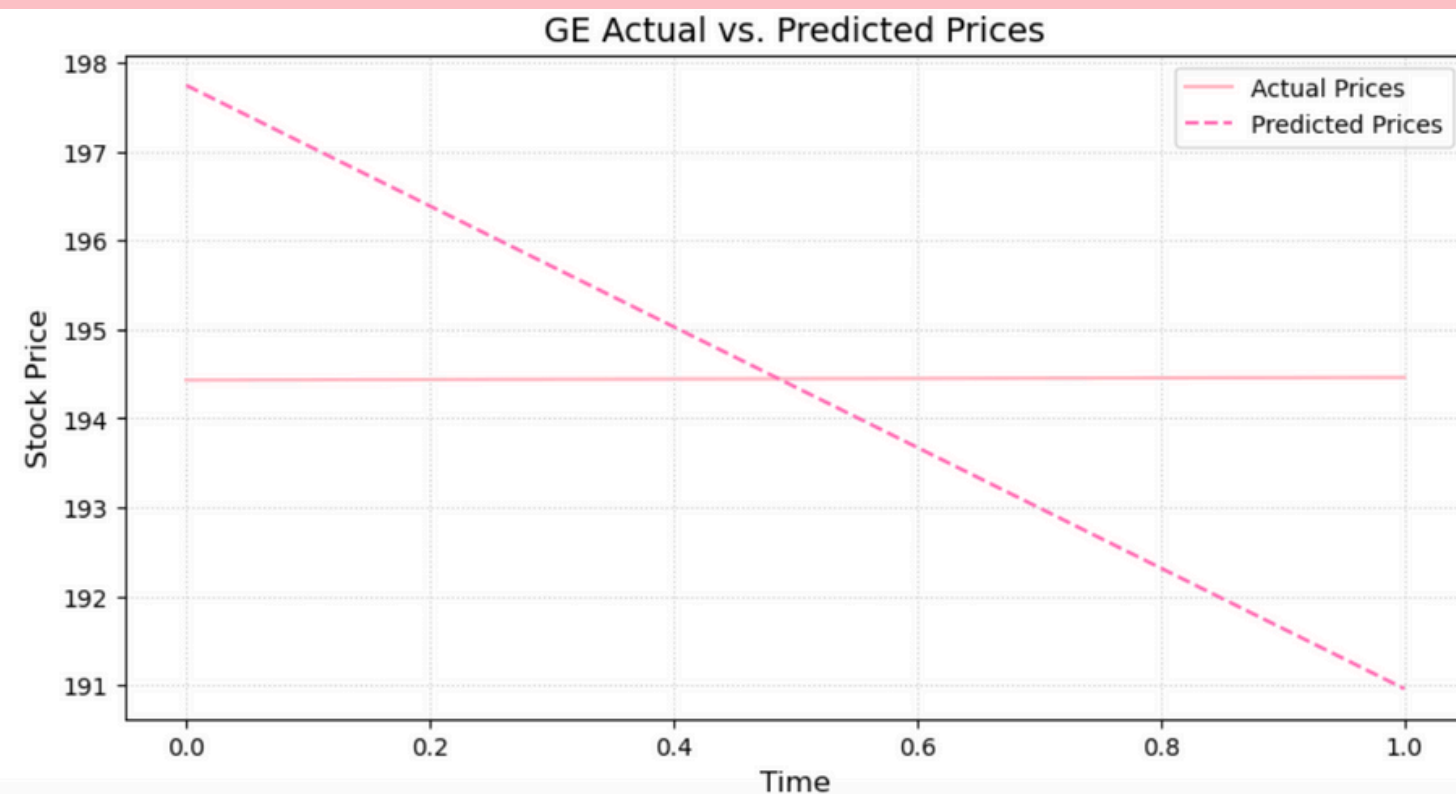
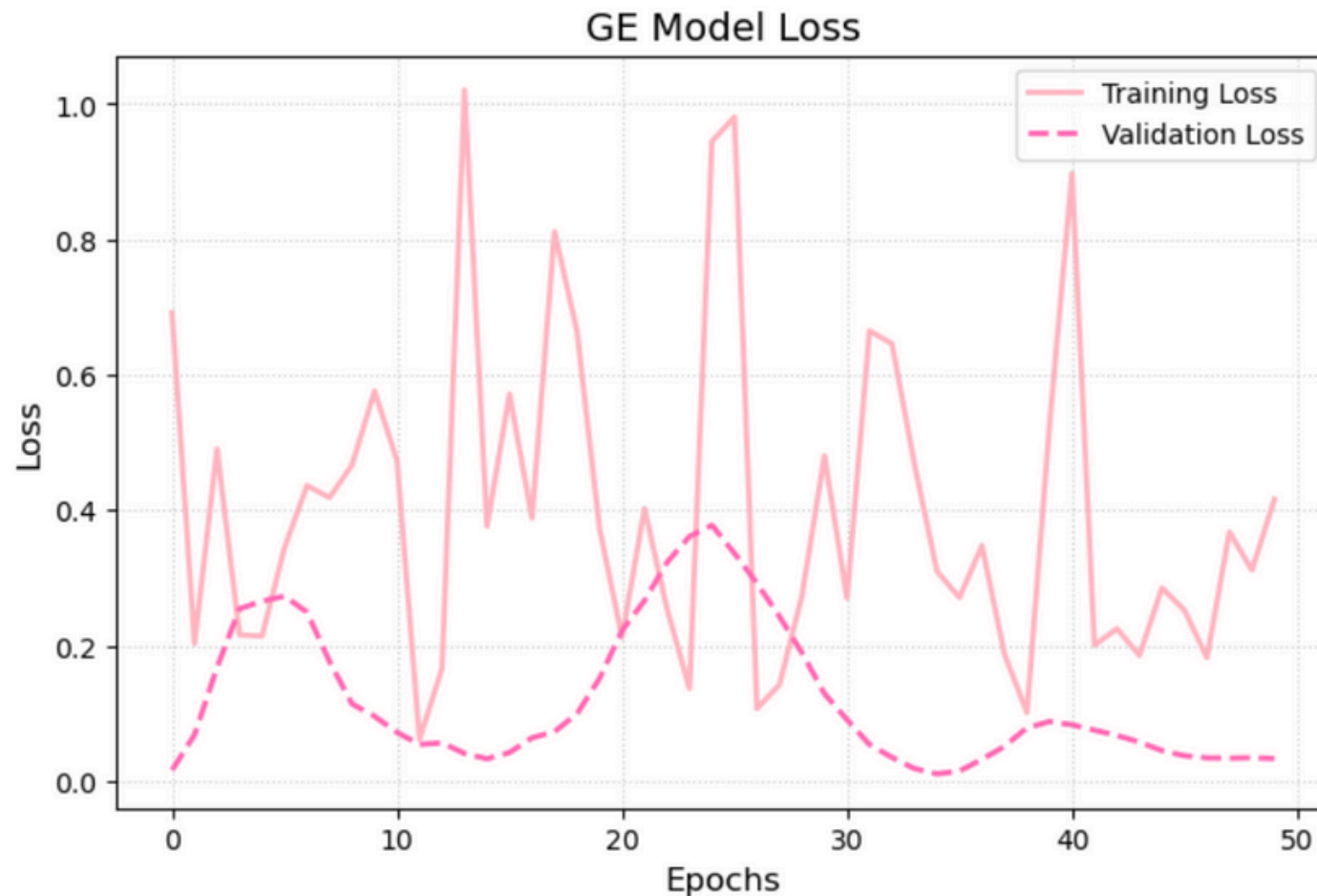
time_steps = 25
X, y = create_sequences(df_scaled, time_steps)
X_train, X_test, y_train, y_test = X[:int(len(X)*0.8)], X[int(len(X)*0.8):], y[:int(len(y)*0.8)], y[int(len(y)*0.8):]

# Build RNN model
model = Sequential([
    SimpleRNN(50, return_sequences=True, input_shape=(time_steps, 1)),
    Dropout(0.4),
    SimpleRNN(50, return_sequences=False),
    Dropout(0.4),
    Dense(1)
])

# Compile model
model.compile(optimizer='adam', loss='mean_squared_error')

# Train model
history = model.fit(X_train, y_train, epochs=50, batch_size=64, validation_data=(X_test, y_test))
```

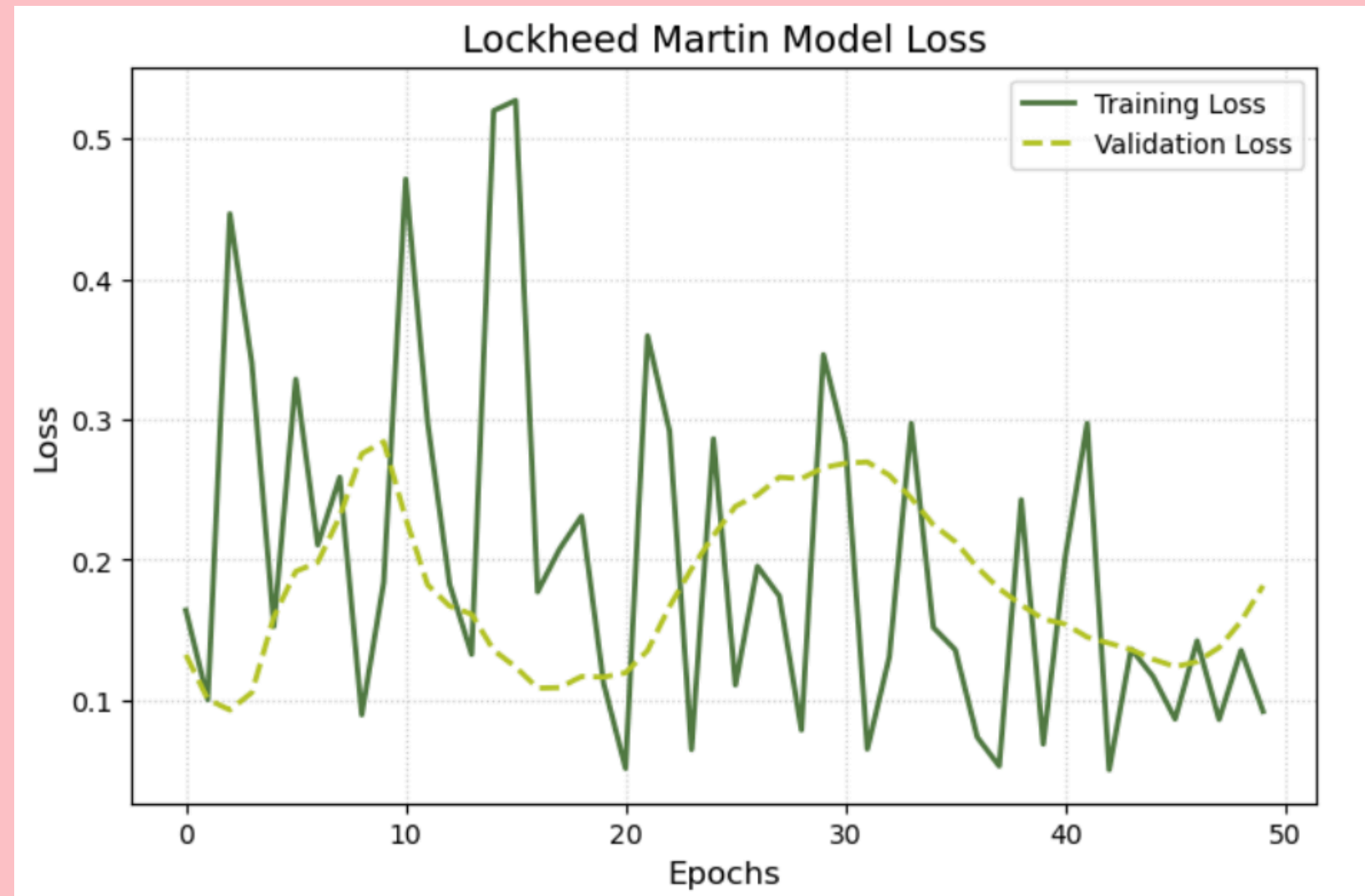
GE Aerospace



Predicted March 13th closing price: \$190.96.

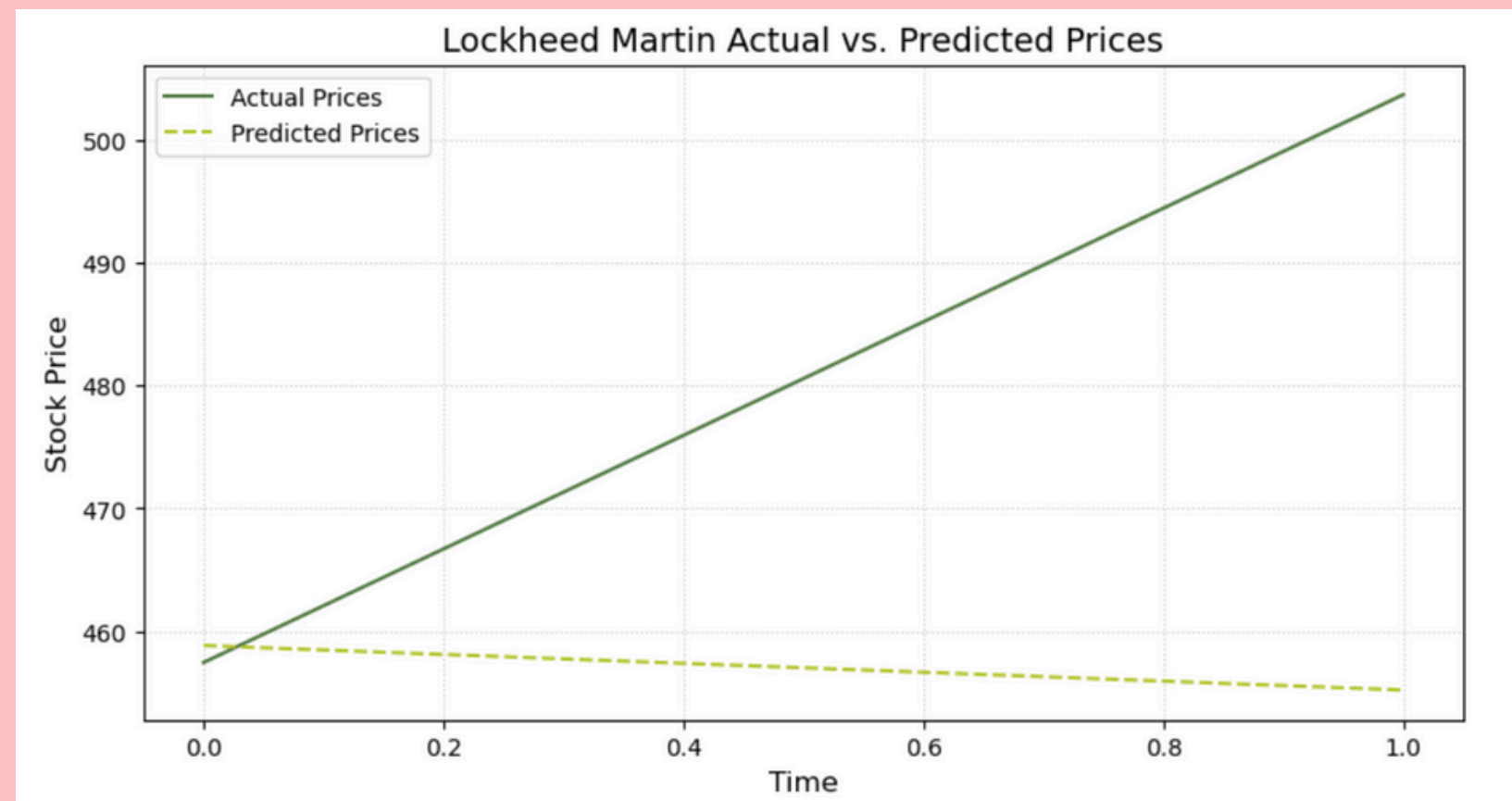
- High RMSE and MAE (3.41), indicating possible overfitting.
- Training loss shows fluctuation, and predicted prices deviate from actuals.
- Suggests volatility issues with the model's generalization.

Lockheed Martin



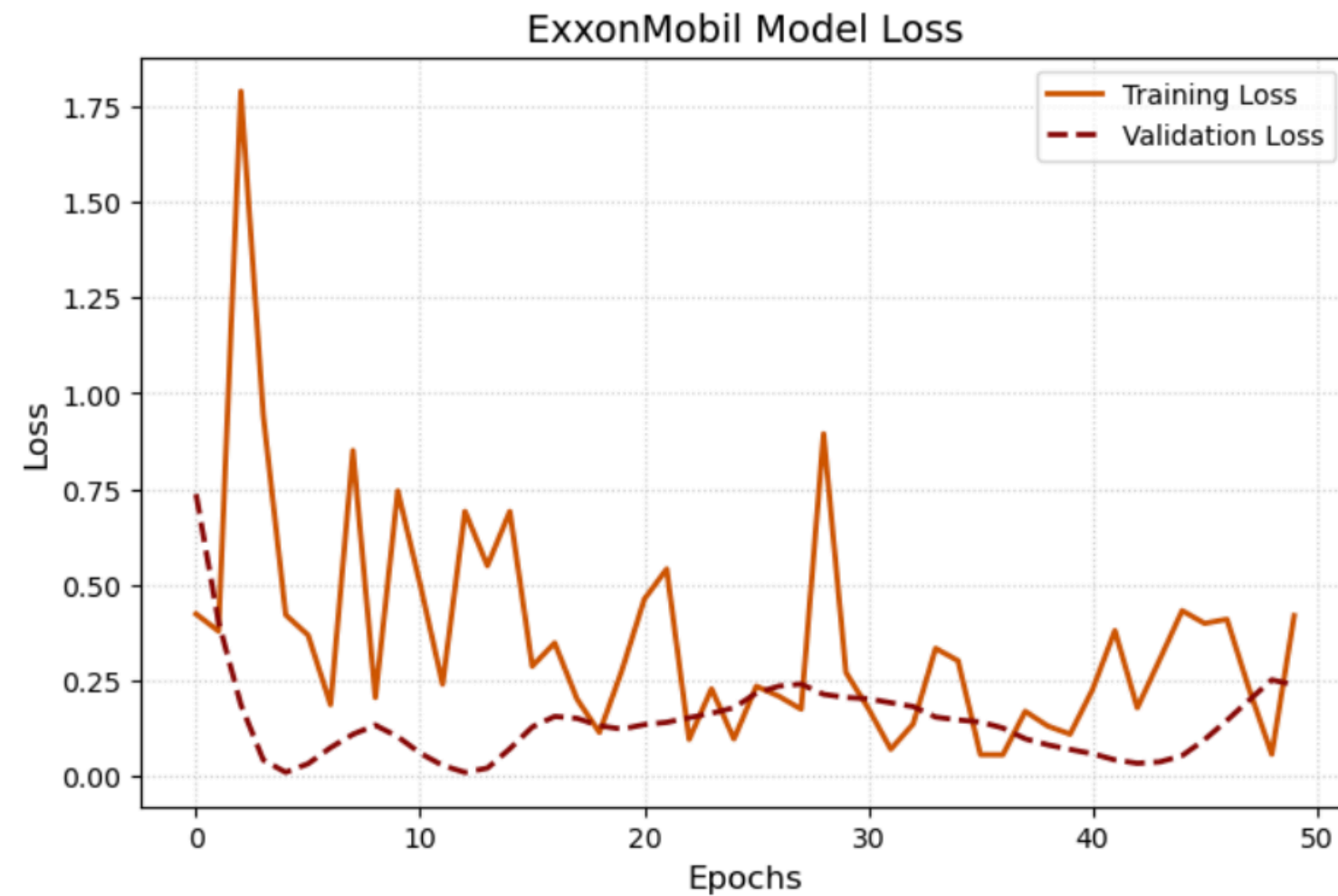
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**Predicted March 13th closing price:
\$455.21.**



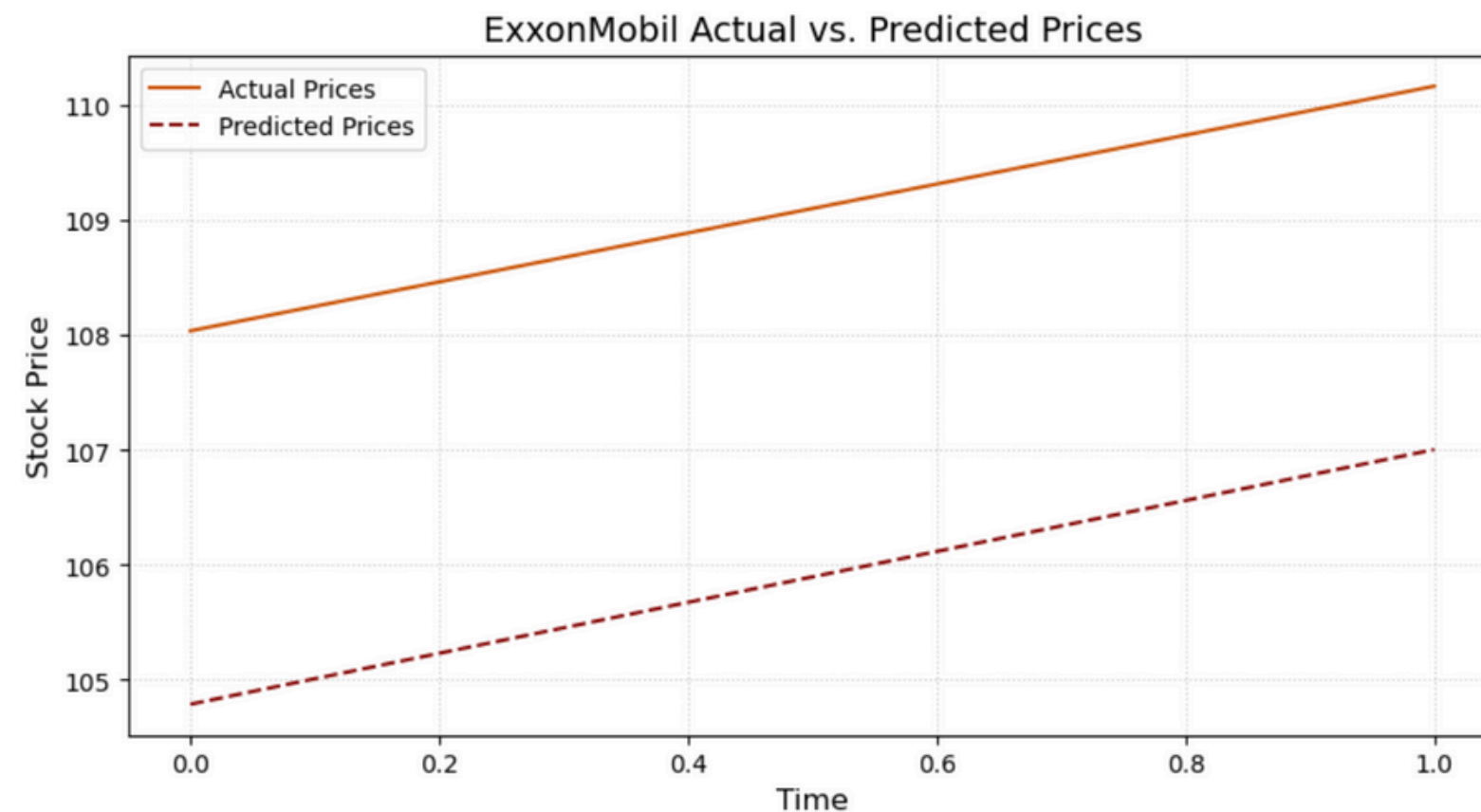
- RMSE and MAE: 34.30 and 24.93, respectively (higher than GE).
- Predicted prices diverge from actuals, indicating inability to capture upward momentum.
- RNN may not be suited for Lockheed's stock behavior.

ExxonMobil



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**Predicted March 13th closing price:
\$107.01.**



- RMSE and MAE: 3.21, indicating good model performance.
- Predicted and actual prices align well, moving upward in parallel.
- Model better captures predictable trends in the energy sector.

Final Thoughts & Future Improvements

- Model would perform better for more stable industries.
 - (But what industry doesn't expect volatility?)
 - Key limitation: sensitivity to volatile price swings
- Future improvements:
 - Incorporate volume as a feature.
 - Explore more complex RNN architectures (e.g., LSTM).
 - Include macroeconomic indicators for better predictive power.

THANKS <3